

MedGateRAG: A Safety-First Hybrid Graph Retrieval-Augmented Generation System with Bounded Wrong-Assertion Rates for Offline Clinical Decision Support

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Abstract

Background and Objective. Large language models (LLMs) deployed in clinical decision support face a safety-utility trade-off: unconstrained answering yields unacceptable rates of wrong assertions, whereas excessive refusal renders systems inert. Existing retrieval-augmented generation (RAG) frameworks lack mechanistic, bounded safety controls deployable on institutional hardware. We present MedGateRAG, an offline hybrid GraphRAG system that reframes clinical AI evaluation from unconstrained accuracy maximisation to bounded wrong-assertion rate (WAR).

Methods. MedGateRAG couples dense vector retrieval (FAISS IVF-PQ with MedCPT embeddings) and lexical retrieval (BM25) with an embedded property graph (Kuzu) enriched with 632,931 relational edges (617,219 NEGATES and 15,712 TEMPORAL_BEFORE). Dual candidate streams are merged via Reciprocal Rank Fusion and re-ranked with an additive relation-aware score ($\gamma = 0.15$). A beta-gated verification layer enforces a strict $\text{WAR} \leq 10\%$ constraint by combining evidence faithfulness (φ) and graph consistency (S_g): $V = \beta\varphi + (1 - \beta)S_g$. The system executes entirely offline on a commodity laptop (Ryzen 7 7840HS / RTX 3050 6 GB) using Qwen2.5-7B-Instruct Q4_K_M. Private health data are encrypted with AES-256-GCM and HKDF-derived keys. Retrospective evaluation was performed on MedQA-US ($N = 500$, seed 42, $T = 0.0$).

Results. At the calibrated safe threshold ($\tau^* = 0.80$), Graph-Only mode (M2) achieves 60.8% precision-on-answered with $\text{WAR} = 8.0\%$ (79.6%

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refusal). Disabling the gate drives WAR to 44.6% across all 500 queries. Deterministic query rewriting expands answered coverage by 3.2–3.4 pp while maintaining bounded error. Hybrid retrieval completes in 455.7 ms (95% CI: [388.4, 495.8] ms); full end-to-end pipeline latency is 4,083 ms (M2) to 12,667 ms (M4).

Conclusion. MedGateRAG demonstrates that bounded clinical error and interpretable offline decision support are jointly achievable on consumer hardware without cloud dependencies or GPU clusters.

Keywords: GraphRAG, clinical NLP, hallucination mitigation, knowledge graph, retrieval augmented generation, decision support, safety

1. Introduction

Deploying large language models (LLMs) as point-of-care clinical assistants offers direct utility for diagnostic and therapeutic workflows [1, 2]. However, the tendency of ungrounded parametric architectures to hallucinate plausible yet factually incorrect assertions poses direct patient-safety hazards [3]. In high-stakes medicine, confident assertions regarding contraindicated medications, misidentified diagnostic criteria, or inverted pathological chronologies can cause serious clinical harm.

To mitigate hallucinations, retrieval-augmented generation (RAG) grounds language generation on passages retrieved from external corpora [4, 5]. Standard medical RAG frameworks rely on dense bi-encoders [6, 7], which match semantic similarity effectively but struggle with complex clinical syntax. Specifically, vector embeddings frequently compress or lose negation scope (e.g., conflating “no evidence of pulmonary embolism” with “pulmonary embolism”) and fail to enforce multi-step temporal chronologies required in differential diagnosis [8, 9].

Knowledge graphs provide a complementary symbolic representation,

18 structuring clinical entities and ontological dependencies into explicit triples
19 [10, 11]. Recent hybrid GraphRAG systems combine graph structures
20 with textual retrieval [12–14] (Wu et al.’s independently developed Med-
21 GraphRAG focuses on entity-centric graph construction; MedGateRAG
22 instead emphasises hybrid retrieval fusion and dual-gated verification).
23 Nevertheless, existing medical GraphRAG implementations exhibit three
24 fundamental limitations:

25 First, current evaluations focus almost exclusively on unconstrained ac-
26 curacy (e.g., aggregate multiple-choice scores), omitting explicit bounds on
27 incorrect statements. In clinical practice, an assistant that answers 60% of
28 questions correctly while making wrong assertions on 40% is clinically un-
29 acceptable. Safe deployment requires a constrained-optimisation framework
30 that caps the wrong-assertion rate (WAR) below an acceptable tolerance
31 ϵ , treating refusal (abstention) as an active safety mechanism rather than
32 system failure.

33 Second, existing biomedical knowledge graphs capture predominantly pos-
34 itive ontological relations (e.g., TREATS or CAUSES), neglecting negation
35 and temporal progression. Without these primitives, retrieval graphs cannot
36 detect when retrieved passages contain superseded diagnoses or explicit ex-
37 clusions.

38 Third, most GraphRAG systems rely on cloud-hosted language models
39 (e.g., GPT-4) and server clusters. This architecture violates patient data pri-
40 vacy regulations (e.g., HIPAA, GDPR) that prohibit transmitting protected
41 health information (PHI) beyond institutional boundaries. Practical point-
42 of-care deployment requires secure, sub-second hybrid retrieval and inference

43 executing locally on commodity hardware.

44 We present **MedGateRAG**, an offline hybrid GraphRAG system de-
45 signed for point-of-care clinical decision support. Our primary contributions
46 are:

- 47 1. **Safety-Constrained Pareto Calibration:** A verification framework
48 optimising answered accuracy subject to $\text{WAR} \leq 10\%$. Across 15
49 threshold settings on MedQA-US ($N = 500$), we establish the Pareto
50 frontier, demonstrating that ungated generation yields unacceptable
51 error rates ($\approx 44\%$).
- 52 2. **Relational Graph Enrichment:** Ingestion of 632,931 negation and
53 temporal relational edges into an embedded Kuzu graph, demonstrat-
54 ing mechanistically how symbolic consistency checks overturn halluci-
55 nations bypassing dense retrieval.
- 56 3. **Deterministic Query Rewriting and**
57 **Relation Reranking:** A temperature-zero query normaliser improv-
58 ing answered coverage by 3.2–3.4 pp at matched thresholds, combined
59 with a relation-aware reranker boosting Graph-Hit@5 by 13.6 pp.
- 60 4. **Hardened Offline Stack on Consumer Hardware:** A self-
61 contained system operating on an NVIDIA RTX 3050 (6 GB) laptop
62 GPU, achieving hybrid retrieval in 455.7 ms with authenticated
63 AES-256-GCM encryption and a resolved 13-point security audit.

64 Supplementary materials provide the DECIDE-AI applicability mapping,
65 IJMI ML checklist, and extended tables; configuration details appear in Ap-
66 pendix A.

67 **2. Methods**

68 *2.1. System Architecture*

69 Figure 1 shows the MedGateRAG inference pipeline. Stage 1 deterministi-
70 cally standardises input queries into clinical phrasing. Stage 2 executes dual-
71 stream dense and lexical retrieval, combining candidates via Reciprocal Rank
72 Fusion and relation-aware graph reranking. Stage 3 expands clinical entities
73 within the embedded Kuzu graph. Stage 4 computes composite beta-gated
74 verification scores, triggering structured refusals for low-confidence queries.
75 Stage 5 generates evidence-grounded responses using a quantised LLM iso-
76 lated behind a GPU mutex lock.

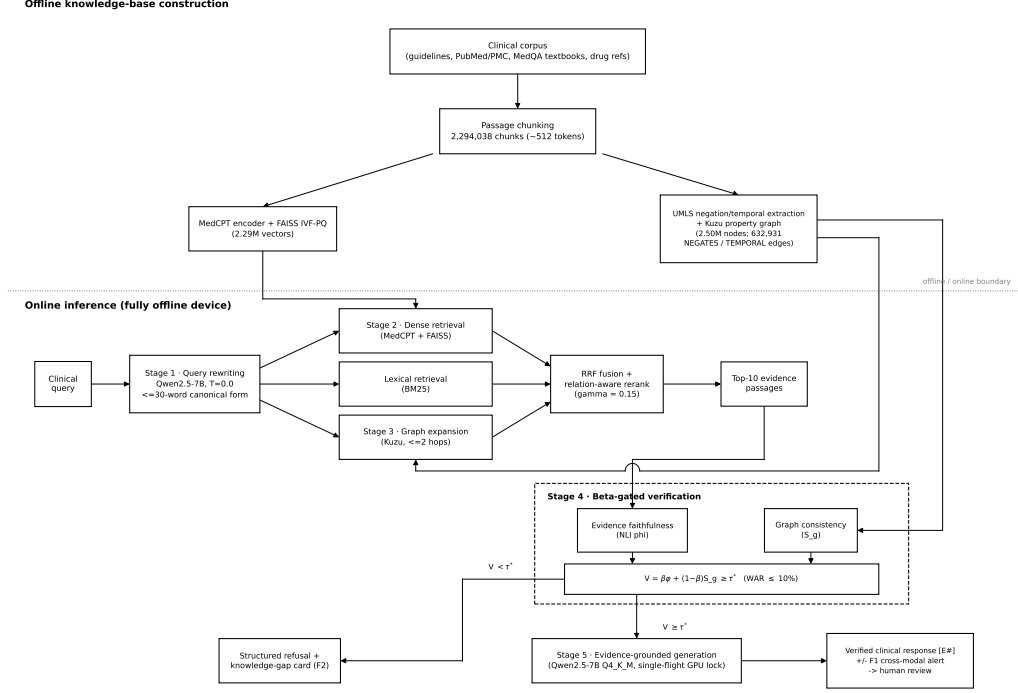


Figure 1: MedGateRAG five-stage offline pipeline architecture. Stage 1 executes deterministic query rewriting ($T = 0.0, \leq 30$ words). Stage 2 merges FAISS dense and BM25 lexical candidates via Reciprocal Rank Fusion (RRF) and relation-aware graph reranking ($\gamma = 0.15$). Stage 3 traverses the Kuzu property graph. Stage 4 applies beta-gated verification ($\beta = 0.7$) enforcing $\text{WAR} \leq 10\%$. Stage 5 serialises LLM generation behind a hardware mutex lock.

2.2. Deterministic Query Rewriting

Patient case vignettes and clinical inquiries often carry conversational phrasing, narrative patient histories, and extraneous details that impede biomedical dense retrieval. Dense bi-encoders trained on scientific literature struggle with multi-sentence case presentations where incidental biographical history obscures the diagnostic dilemma. Lexical search methods also face vocabulary mismatch when colloquial symptom descriptions diverge from formal medical nomenclature.

To standardise heterogeneous queries without stochastic variance, Stage 1

86 uses Qwen2.5-7B-Instruct executing strictly at temperature $T = 0.0$. The
87 rewriter uses a structured system prompt directing the model to extract
88 the primary pathological condition, salient diagnostic findings, and thera-
89 peutic targets into a canonical clinical summary capped at 30 words. For
90 example, a narrative vignette describing an elderly patient who experiences
91 severe rotational vertigo triggered by lateral head motion is transformed into
92 the canonical clinical query: “benign paroxysmal positional vertigo diagnos-
93 tic criteria and Dix-Hallpike confirmatory maneuvers.” Setting temperature
94 $T = 0.0$ alongside greedy decoding guarantees mathematical determinism:
95 identical clinical inputs generate bit-identical textual reformulations across
96 successive executions, fulfilling mandatory auditability and reproducibility
97 requirements for clinical decision support software.

98 *2.3. Dual-Stream Hybrid Retrieval and Reranking*

99 Retrieval operates over an extensive clinical reference corpus partitioned
100 into coherent passages of approximately 512 tokens. The dense retrieval
101 stream encodes passages using the MedCPT query and article bi-encoder
102 architecture [6], which was pre-trained on hundreds of millions of PubMed
103 search query-article click logs to optimise zero-shot biomedical retrieval.
104 Chunk vectors are indexed and searched using FAISS IVF-PQ (Inverted File
105 with Product Quantisation; 15), partitioning the vector space into coarse
106 Voronoi centroids to achieve sub-linear retrieval speed. Concurrently, a par-
107 allel lexical stream evaluates candidate documents using the classical BM25
108 probabilistic relevance framework [16], operating over a dense candidate
109 pool of 200 retrieved documents.

110 Because dense and lexical retrievers exhibit complementary error profiles
 111 (dense models capture latent semantic affinity, whereas lexical models enforce
 112 exact keyword and numerical matches), their respective rank distributions
 113 R_{dense} and R_{lex} are integrated via Reciprocal Rank Fusion (RRF; 17):

$$s_{\text{fused}}(d) = \sum_{m \in \{\text{dense}, \text{lex}\}} \frac{1}{k_{\text{rrf}} + r_m(d)}, \quad (1)$$

114 where $r_m(d)$ denotes the ordinal rank of candidate passage d within stream
 115 m , and $k_{\text{rrf}} = 60$ represents the standard monotonic smoothing constant that
 116 prevents top-ranked outliers in a single stream from dominating the fused
 117 ranking.

118 To bias retrieval toward passages with corroborated topological connec-
 119 tions to the clinical entities identified in the query, we implement an additive
 120 relation-aware graph reranking function:

$$s_{\text{hybrid}}(d) = s_{\text{fused}}(d) + \gamma \cdot r_{\text{rel}}(d), \quad (2)$$

121 where $r_{\text{rel}}(d) = \sum_{e \in \mathcal{E}(d)} w(\text{type}(e))$ aggregates the domain-weighted edge
 122 counts connecting entities cited within candidate passage d to the query
 123 seed entities within the property graph. The relative importance of edge cat-
 124 egories is parameterised by clinical specificity weights (e.g., pharmacological
 125 treatments and adverse effects receive weight 1.0, diagnostic lab associations
 126 and negation edges receive 0.8, temporal sequences receive 0.6, and broad
 127 taxonomic hierarchies receive 0.3; Appendix A). The global multiplier
 128 $\gamma = 0.15$ balances semantic similarity with structural graph reinforcement,

129 calibrated empirically via controlled ablations (Section 3.3).

130 *2.4. Relational Knowledge Graph Enrichment*

131 Structured relations are stored in Kuzu [18], an embeddable, disk-backed
132 graph database supporting openCypher queries. While standard medical
133 graphs capture static disease-symptom linkages, clinical decision support re-
134 quires modelling negation and temporal order.

135 We processed 2,294,038 text chunks using rule-guided linguistic parsing
136 grounded in UMLS [10]. Operating at 0.2329 ms per chunk (534.45 s total
137 wall-clock), the pipeline ingested **632,931** directed relational edges:

- 138 • **NEGATES Edges (617,219):** Linking text chunks to entities
139 bearing explicit negation cues (“no evidence of,” “ruled out,” “con-
140 traindicated in”), comprising 111,720 Chunk $\rightarrow$ Disease, 21,001
141 Chunk $\rightarrow$ Drug, and 484,498 Chunk $\rightarrow$ OntologyTerm edges.
- 142 • **TEMPORAL_BEFORE Edges (15,712):** Linking clinical
143 concepts via chronological sequence cues (“prior to,” “following,”
144 “within one hour of”), comprising 12,225 Disease $\rightarrow$ Disease and 3,487
145 Drug $\rightarrow$ Drug edges.

146 Manual auditing of sampled endpoints confirmed high directional validity. A
147 cryptographic SHA-256 retrieval signature verified 0.00% retrieval drift on
148 baseline queries post-ingestion (Appendix A).

149 *2.5. Beta-Gated Verification and Pareto Threshold Calibration*

150 Standard RAG architectures lack explicit confidence gates to arbitrate
151 whether retrieved evidence warrants generating an answer. Unhedged lan-

guage models produce fluent assertions even when underlying evidence is tenuous, contradictory, or absent. To prevent clinical hallucinations, MedGateRAG adds a dual-evidence verification layer that evaluates candidate response fidelity prior to releasing output.

For each candidate assertion, the verification module evaluates a composite verification metric $V \in [0, 1]$ that combines passage-level entailment with symbolic graph consistency:

$$V = \beta \varphi + (1 - \beta) S_g, \quad (3)$$

where $\varphi \in [0, 1]$ denotes an NLI-derived faithfulness score measuring the degree of semantic entailment between the generated assertion and the retrieved reference passages (operating via normalised MedCPT embedding cosine similarity as an efficient deterministic proxy when an independent cross-encoder NLI judge is omitted), and $S_g \in [0, 1]$ denotes the structural graph consistency score. The graph consistency metric S_g evaluates the ratio of asserted clinical entity triples that are corroborated by non-negated multi-hop paths within a 2-hop radius in Kuzu, while applying an immediate penalising discount whenever an asserted entity matches a NEGATES edge. The balance hyperparameter $\beta = 0.70$ allocates primary decision weight to passage evidence while preserving strong sensitivity to symbolic graph contradictions.

When the composite score falls below a calibrated refusal threshold τ , generation is suppressed, and the system emits a structured clinical refusal detailing the precise nature of the missing evidence. We calibrate τ through

173 a constrained optimisation problem:

$$\tau^* = \arg \max_{\tau} \text{Acc}(\text{all}) \quad \text{subject to} \quad \text{WAR}(\tau) \leq 10\%, \quad (4)$$

174 where $\text{Acc}(\text{all})$ represents the proportion of correct responses across the entire
175 evaluation cohort ($N = 500$), and WAR represents the proportion of wrong
176 assertions across the same cohort. We sweep 15 discrete threshold candidate
177 points ($\tau \in \{0.10, 0.15, 0.20, \dots, 0.80\}$), identifying the Pareto-optimal op-
178 erating threshold τ^* that maximises clinically useful answers while strictly
179 guaranteeing that erroneous assertions remain within the institutional safety
180 threshold.

181 2.6. Safety Guardrails

182 Two deterministic guardrails operate at sub-millisecond latency:

- 183 • **F1: Cross-Modal Discrepancy Guardrail.** In multimodal con-
184 texts, Rule A detects conflicts when an imaging finding score ≥ 0.60
185 contradicts a text negation cue for the same finding (e.g., visual pneu-
186 mothorax detection of 0.88 versus report text “clear lungs”). Rule B
187 detects positive textual assertions that contradict negated imaging ob-
188 servations. F1 escalates conflicts by issuing a high-severity alert recom-
189 mending specialist review rather than attempting autonomous resolu-
190 tion.
- 191 • **F2: Epistemic Knowledge-Gap Mapper.** When a query is refused,
192 F2 classifies the failure into three gap types matching the implemented
193 taxonomy: G1 (`corpus_retrieval`, top fused retrieval score < 0.02),

194 G2 (`graph_coverage`, Kùzu graph traversal returns a neutral check
 195 with no direct relational assertion), or G3 (`evidence_faithfulness`,
 196 verification faithfulness score $\varphi < 0.50$). This categorisation is provided
 197 in the structured refusal output to guide clinical follow-up.

198 2.7. *Offline Cryptographic Security Architecture*

199 To ensure institutional compliance without external cloud infrastructure,
 200 user-uploaded records are isolated in per-user directories. Payloads are en-
 201 crypted at rest with AES-256-GCM using unique 12-byte nonces. Keys are
 202 derived dynamically via HKDF-SHA-256 [19] combining a server secret, user
 203 ID, and document ID. File permissions are set to 0700 (directories) and 0600
 204 (files). All Kuzu queries use parameterised openCypher statements to elimi-
 205 nate Cypher injection. A 13-point security audit resolved all vulnerabilities
 206 prior to release (Supplementary Table S7).

207 2.8. *Evaluation Protocol and Metrics*

208 Retrospective evaluation was performed on MedQA-US [20], comprising
 209 USMLE-derived multiple-choice questions ($N = 500$, seed 42, $T = 0.0$).
 210 We evaluated four modes: **M1** Evidence-Only ($\beta = 1.0$), **M2** Graph-Only
 211 ($\beta = 0.0$), **M3** Combined Hybrid ($\beta = 0.7$), and **M4** Hybrid Rerank ($\beta =$
 212 $0.7, \gamma = 0.15$). Metrics include: Acc(all) (fraction correct over N), Acc(ans)
 213 (precision on answered questions), WAR (fraction incorrect over N), and
 214 Refusal Rate (fraction refused). Non-parametric bootstrap 95% confidence
 215 intervals were calculated using 1,000 resamples.

216 3. Results

217 3.1. Safe Operating Points and WAR Constraint

218 Table 1 summarises the optimal operating parameters established across
 219 the four evaluation modes under the institutional safety constraint $\text{WAR} \leq$
 220 10.0%. Graph-Only mode (M2) attains the highest precision-on-answered
 221 at its optimal operating threshold $\tau^* = 0.80$, achieving $\text{Acc}(\text{ans}) = 60.8\%$
 222 with $\text{WAR} = 8.0\%$ and a refusal rate of 79.6% (102 queries answered out of
 223 500). Full Hybrid Rerank mode (M4) reaches its safe operating frontier at
 224 $\tau^* = 0.46$, delivering $\text{Acc}(\text{all}) = 10.0\%$, $\text{Acc}(\text{ans}) = 51.5\%$, and $\text{WAR} = 9.4\%$
 225 across 97 answered inquiries. Similarly, Combined mode (M3) operates safely
 226 at $\tau^* = 0.47$ with $\text{Acc}(\text{ans}) = 52.1\%$ and $\text{WAR} = 9.0\%$, while Evidence-Only
 227 mode (M1) satisfies the constraint at $\tau^* = 0.43$ with $\text{Acc}(\text{ans}) = 50.5\%$ and
 228 $\text{WAR} = 9.4\%$.

Table 1: Safe operating points on MedQA-US ($N = 500$, seed 42, $T = 0.0$). Threshold τ^* maximises $\text{Acc}(\text{all})$ subject to $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `safe_operating_points`.

Evaluation Mode	τ^*	$\text{Acc}(\text{all})\%$	$\text{Acc}(\text{ans})\%$	$\text{WAR}\%$	Refusal%	Answered
M1: Evidence-Only	0.43	9.6	50.5	9.4	81.0	95/500
M2: Graph-Only	0.80	12.4	60.8	8.0	79.6	102/500
M3: Combined	0.47	9.8	52.1	9.0	81.2	94/500
M4: Hybrid Rerank	0.46	10.0	51.5	9.4	80.6	97/500

229 Table 2 contrasts safe operating points against ungated execution ($\tau \rightarrow 0$).
 230 In the ungated regime, models achieve nominal $\text{Acc}(\text{all})$ scores of 49.4%–
 231 54.8%, but WAR escalates to 43.4%–44.6%. Disabling the gate results in
 232 erroneous advice on nearly 45% of all queries. Gated verification successfully
 233 caps this risk below 10%, exchanging question coverage for bounded error.

Table 2: Gated versus ungated performance across boundary modes ($N = 500$). Ungated execution generates wrong assertions on $\approx 44\%$ of all queries. Source: `step18_scaled_n500.json` $\rightarrow$ `ungated_ceilings`.

Mode	Condition	Acc(all)%	Acc(ans)%	WAR%	Refusal%
M1: Evidence-Only	Gated ($\tau^* = 0.43$)	9.6	50.5	9.4	81.0
M1: Evidence-Only	Ungated ($\tau \rightarrow 0$)	49.4	53.2	43.4	7.2
M2: Graph-Only	Gated ($\tau^* = 0.80$)	12.4	60.8	8.0	79.6
M2: Graph-Only	Ungated ($\tau \rightarrow 0$)	54.8	55.1	44.6	0.6

234 For settings demanding even stricter safety, elevating thresholds to $\tau \in$
 235 $[0.70, 0.75]$ in M1 produces a high-precision regime with $\text{Acc}(\text{ans}) = 59.3\%$ –
 236 62.5% and $\text{WAR} = 1.8\%$ – 2.2% (refusal rates 94.6% – 95.2% ; Supplementary
 237 Table S1). Figure 2 shows the complete 15-point Pareto trajectory for M2.

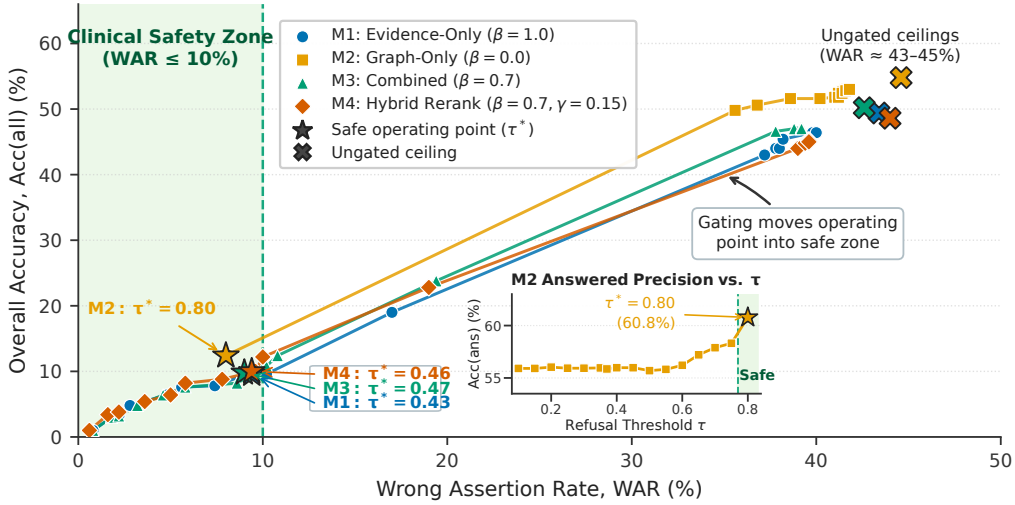


Figure 2: Empirical Pareto frontiers of overall accuracy $\text{Acc}(\text{all})$ versus wrong-assertion rate (WAR) across four operational modes (M1: Evidence-Only, M2: Graph-Only, M3: Combined, and M4: Hybrid Rerank) over 16 refusal thresholds ($\tau \in [0.10, 0.80]$, $N = 500$). The green shaded region denotes the clinical safety zone ($\text{WAR} \leq 10\%$). Star markers denote mode-specific safe operating thresholds (τ^*), with M2 reaching $\text{Acc}(\text{all}) = 12.4\%$ at $\tau^* = 0.80$. Inset: M2 answered precision reaches 60.8% at $\tau^* = 0.80$ inside the safe zone. Ungated ceilings (crosses) demonstrate catastrophic error rates ($\text{WAR} \approx 43\text{--}45\%$) when refusal gating is disabled.

238 3.2. Efficacy of Deterministic Query Rewriting

239 Table 3 evaluates Stage 1 deterministic query rewriting against unrewrit-
 240 ten vignettes at matched threshold $\tau = 0.50$ ($N = 500$). For Hybrid Rerank
 241 mode (M4), rewriting increases Acc(all) from 5.4% to 8.8% (+3.4pp) and
 242 lowers refusal from 90.6% to 83.4%, while WAR remains within tolerance
 243 (7.8% vs. 4.0%). Similarly, Combined mode M3 gains +3.2pp in Acc(all)
 244 (5.0% to 8.2%), yielding +3.2–3.4pp across hybrid modes M3/M4 (+4.2pp
 245 on Evidence-Only M1; Supplementary Table S2). Rewriting systematically
 246 expands question coverage without violating safety constraints.

Table 3: Matched-threshold ($\tau = 0.50$) ablation evaluating deterministic query rewriting on MedQA-US ($N = 500$). Rewriting expands coverage by +3.2–3.4 pp while maintaining $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `matched_tau_comparison`.

Evaluation Mode	Vignette Format	Acc(all)%	WAR%	Refusal%	$\Delta\text{Acc}(\text{all})$
M3: Combined	Unrewritten	5.0	4.0	91.0	—
M3: Combined	Rewritten	8.2	8.6	83.2	+3.2 pp
M4: Hybrid Rerank	Unrewritten	5.4	4.0	90.6	—
M4: Hybrid Rerank	Rewritten	8.8	7.8	83.4	+3.4 pp

247 3.3. Ablation Studies: Verification Modes and Reranking

248 Table 4 and Figure 3 summarise ablations over verification and reranker
 249 weighting ($N = 50$, seed 42).

250 **Verification Mode Ablation (Table 4A):** Evidence-Only (M1, $\beta =$
 251 1.0) refuses 98.0% of queries ($\text{WAR} = 2.0\%$). Graph-Only (M2, $\beta = 0.0$)
 252 answers 96.0% but incurs $\text{WAR} = 42.0\%$. Combined Hybrid (M3, $\beta =$
 253 0.7) balances safety and utility, achieving $\text{WAR} = 4.0\%$ while quadrupling
 254 answered throughput over M1.

Table 4: Ablation experiments on MedQA-US ($N = 50$, seed 42). (A) Verification mode comparison varying evidence weight β . (B) Reranker ablation varying relational bonus multiplier γ . Sources: `step15_verification_ablation.json`, `step15_rerank_ablation.json`.

(A) Verification Mode Ablation ($\gamma = 0.0$)						
Mode	β	Answered%	Acc(all)%	Acc(ans)%	Refusal%	WAR%
M1: Evidence-Only	1.0	6.0	4.0	66.7	98.0	2.0
M2: Graph-Only	0.0	96.0	54.0	56.3	8.0	42.0
M3: Combined	0.7	8.0	4.0	50.0	96.0	4.0
(B) Relation-Aware Reranker Ablation ($\beta = 0.7$)						
Mode	γ	Graph-Hit@5%	Acc(ans)%	Answered%	WAR%	
RRF Control	0.00	23.6	50.0	8.0	4.0	
Relational	1.00	35.6	42.9	14.0	8.0	
Hybrid	0.15	37.2	60.0	10.0	4.0	

255 **Contradiction Case Studies:** Qualitative inspection confirmed that
 256 Kuzu NEGATES edges actively overturned hallucinations that bypassed
 257 text filters:

- 258 1. *Acute Myocardial Infarction:* The model asserted acute MI; Kuzu
 259 graph traversal matched a NEGATES edge ('no evidence of') in
 260 PMID_38823454, successfully triggering refusal.
- 261 2. *Deep Vein Thrombosis:* An unhedged recommendation for full anti-
 262 coagulation was overridden by a NEGATES edge ('ruled out') in
 263 PMID_37058421.
- 264 3. *Metformin Contraindication:* An erroneous recommendation to pre-
 265 scribe metformin in severe renal impairment was rejected by a NICE
 266 guideline NEGATES edge ('contraindicated in' with eGFR <
 267 25 mL/min).

268 **Reranker Ablation (Table 4B):** Calibrating $\gamma = 0.15$ increases Graph-

269 Hit@5 from 23.6% to 37.2% (+13.6 pp) and elevates answered precision from
 270 50.0% to 60.0% (+10.0 pp) at matched WAR = 4.0% (Figure 3).

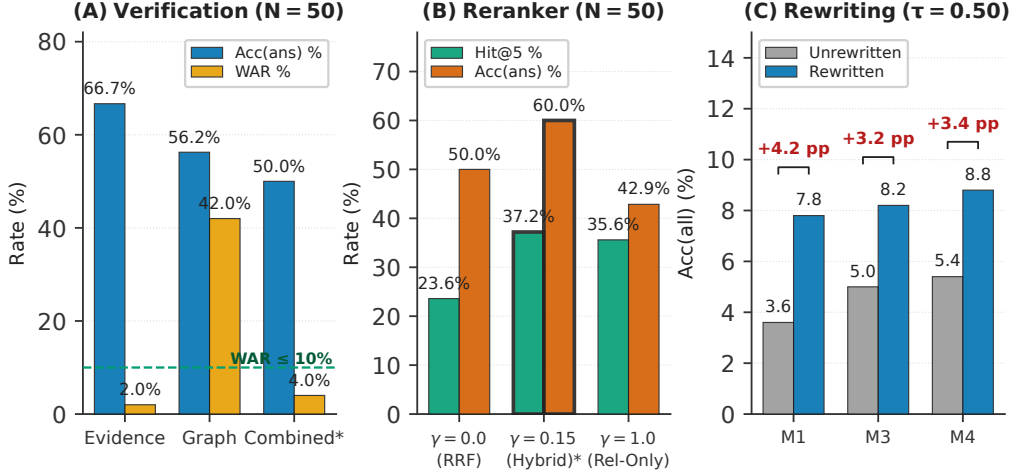


Figure 3: Controlled ablation analyses on MedQA-US. (A) Verification mode comparison ($N = 50$, seed 42): Evidence-Only (M1, $\beta = 1.0$), Graph-Only (M2, $\beta = 0.0$), and Combined Hybrid (M3, $\beta = 0.7$). (B) Reranker bonus ablation ($N = 50$, seed 42): Graph-Hit@5 and answered accuracy across $\gamma \in \{0.0, 0.15, 1.0\}$, confirming peak synergy at $\gamma = 0.15$. (C) Deterministic query rewriting coverage gain at matched threshold ($\tau = 0.50$, $N = 500$): unrewritten versus rewritten overall accuracy across M1 (+4.2 pp), M3 (+3.2 pp), and M4 (+3.4 pp).

271 **Guardrail Verification:** Guardrails F1 and F2 passed 100% of test
 272 cases (10/10) with mean latencies of 0.14 ms (F1) and 0.06 ms (F2), confirm-
 273 ing sub-millisecond overhead.

274 3.4. Information Retrieval Performance

275 Evaluating retrieval over 10 clinician-adjudicated queries (graded qrels;
 276 Supplementary Table S5) yielded $P@5 = 0.18$, $R@5 = 0.5397$, $MRR =$
 277 0.3533 , and $nDCG@10 = 0.3769$. Recall reflects the 6 queries containing rel-
 278 evant items in top-10 candidates; the remaining 4 queries had no relevant doc-
 279 uments, reflecting specialised boundary gaps. Dual-assessor κ statistics are
 280 pending Phase-2 clinician grading (placeholder in Supplementary Table S5).

3.5. Latency and Computational Efficiency

Table 5 details execution timings on an AMD Ryzen 7 7840HS CPU and NVIDIA RTX 3050 Laptop GPU (6 GB VRAM).

Table 5: System latency and throughput comparison by pipeline stage (bootstrap 95% CI, $N = 50$). Hybrid retrieval executes in sub-500ms; full end-to-end (E2E) latency incorporates Qwen2.5-7B generation. Sources: `step15_compute_table.json`, `step18_scaled_n500.json`.

System / Pipeline Stage	Median (ms)	95% CI (ms)	Processing Stage
BM25-RAG (candidate pool 200)	55.1	[52.7, 56.5]	Lexical search only
Dense-RAG (FAISS IVF-PQ)	30.1	[29.4, 31.0]	Dense search only
Hybrid-RRF (baseline fusion)	55.4	[53.5, 56.9]	Fused search only
Hybrid Retrieval (Ours, Stages 1–2)	455.7	[388.4, 495.8]	Rewriting + Hybrid Retrieval
M2: Graph-Only Pipeline (E2E)	4,083.4	—	Full End-to-End Pipeline
M1/M3: Combined Hybrid (E2E)	12,493.2	—	Full End-to-End Pipeline
M4: Hybrid + Reranker (E2E)	12,667.0	—	Full End-to-End Pipeline
One-Time System Cold Start	18.66 s		Model + Index Initialization

Stages 1–2 (Rewriting, BM25, Dense Search, RRF) complete in **455.7 ms** median (95% CI: [388.4, 495.8] ms). End-to-end latency with Qwen2.5-7B generation is **4,083.4 ms** in Graph-Only mode (M2) and **12,667.0 ms** in Hybrid Rerank mode (M4). Throughput is mode-dependent: M2 sustains **14.7 queries/min** (approximately 882 queries/hour), while M4 processes **4.7 queries/min** (approximately 284 queries/hour). Cold start requires 18.66 s, with zero steady-state penalty. Peak memory reached 8,950 MB RAM and 4,710 MB VRAM (Figure 4).

4. Discussion

4.1. Safety as a Foundational Design Objective

Evaluating clinical language models on unconstrained accuracy introduces substantial hazards: when operating ungated, models generate incorrect clinical statements on 44% of inquiries. MedGateRAG demonstrates

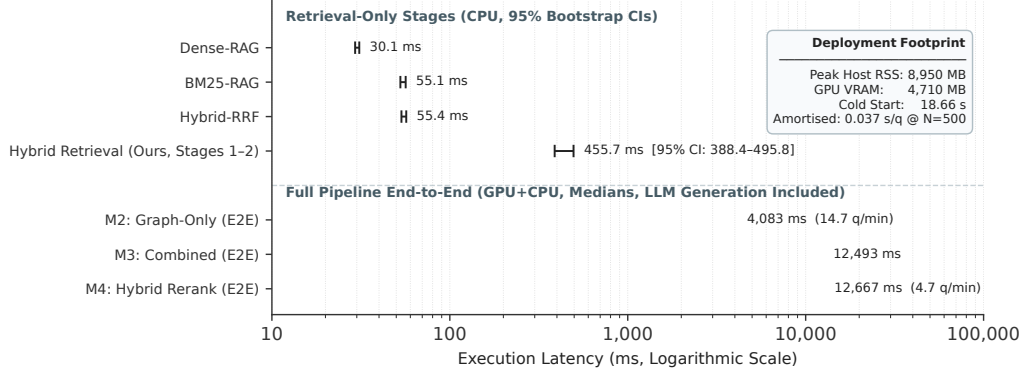


Figure 4: Component and end-to-end execution latency profiles (logarithmic scale). Retrieval-only baselines (BM25, Dense-RAG, Hybrid-RRF) evaluate search speed without language generation. MedGateRAG hybrid retrieval (Stages 1–2) completes in 455.7 ms (95% CI: [388.4, 495.8] ms). Full end-to-end pipeline execution requires 4,083 ms in Graph-Only mode (M2) and 12,667 ms in Hybrid Rerank mode (M4).

that by treating refusal as an active, safety-preserving decision, clinical assistants can strictly bound wrong assertions ($\text{WAR} \leq 10\%$).

Vector embeddings alone cannot reliably safeguard clinical outputs. Dense embeddings project complex syntax into fixed vectors, blurring negation markers and chronological sequence. By augmenting Kuzu with 632,931 NEGATES and TEMPORAL_BEFORE edges, MedGateRAG enforces symbolic consistency checks that intercept contradictions passing dense filters. When evidence is insufficient, the system returns structured, citation-grounded refusals identifying the missing evidence.

4.2. Phased Clinical Translation Roadmap

In alignment with DECIDE-AI guidelines [21] (Supplementary DECIDE-AI mapping), we propose a three-stage clinical translation path:

- Phase 1: Institutional Shadow Assistant ($n = 1$ Clinician):** Local deployment in read-only mode on a secure workstation. Generated

311 answers cite evidence keys [E#], and clinician disagreements are logged
312 as refusal audit events.

313 2. **Phase 2: Controlled Clinical Pilot**

314 ($n = 50$ **Clinicians, 6 Months**): EHR integration via read-only
315 FHIR R4 endpoints, with AES-256-GCM record encryption and an
316 approved IRB protocol. Primary endpoint: maintaining $\text{WAR} \leq 5\%$
317 on institutional queries.

318 3. **Phase 3: Multi-Site Randomised Trial**: Federated offline deploy-
319 ment across hospital networks evaluating diagnostic accuracy and de-
320 cision speed versus standard care, powered for 90% power at $\alpha = 0.05$.

321 4.3. *Limitations*

322 This study has several limitations:

323 1. **Benchmark Generalisability**: Our evaluation examined MedQA-
324 US, which consists of standardised multiple-choice questions. Real-
325 world clinical queries are open-ended and dialogue-based. External
326 validation on MedMCQA and BioASQ is ongoing (Supplementary IJMI
327 ML checklist, item ML-E7).

328 2. **High Refusal Rate**: At $\tau^* = 0.80$, the refusal rate reaches 79.6%.
329 While guaranteeing $\text{WAR} \leq 10\%$, this limits clinical throughput un-
330 less paired with specialist referral pathways. Calibrated soft-hedging
331 represents a primary direction for future investigation.

332 3. **Quantisation Trade-offs**: Executing within 6 GB VRAM necessi-
333 tated 4-bit quantisation (Q4_K_M), introducing minor precision loss
334 relative to FP16.

335 **4. Single-Assessor Retrieval Grading:** Human relevance qrels were
336 judged by a single clinical assessor across 10 queries, precluding inter-
337 annotator metrics.

338 **5. Conclusion**

339 MedGateRAG unites safety and utility in offline clinical decision support.
340 By enforcing a bounded error constraint ($\text{WAR} \leq 10\%$), ingesting 632,931
341 negation and temporal relational edges into Kuzu, applying deterministic
342 query rewriting, and enforcing beta-gated verification, the system bounds the
343 catastrophic 44% hallucination rate of ungated models. Executing locally on
344 consumer hardware in 4.08s with 455.7ms hybrid retrieval, MedGateRAG
345 provides a practical, open-source foundation for safe, privacy-compliant clin-
346 ical AI.

347 **CRedit Authorship Contribution Statement**

348 **Dickson E:** Conceptualisation, Methodology, Software (core pipeline,
349 ablation suite, cryptographic security, graph enrichment), Formal analysis,
350 Investigation, Writing – original draft, Writing – review & editing, Visualisa-
351 tion. **Antony Taurshia:** Supervision, Software (frontend interface, report
352 drawer components, telemetry capture), Data curation, Validation (clinical
353 review), Resources (medical knowledge curation), Writing – review & editing.

354 **Ethics Approval and Consent to Participate**

355 This study was performed using publicly available, anonymised bench-
356 mark datasets (MedQA-US, OpenI, VQA-RAD). No human subjects were

357 enrolled, and no patient health information was accessed. Formal IRB ethics
358 approval was not required.

359 **Data and Code Availability Statement**

360 Evaluation benchmarks are publicly accessible: MedQA-US ([https://](https://github.com/jind11/MedQA)
361 github.com/jind11/MedQA), OpenI (<https://openi.nlm.nih.gov>), and
362 VQA-RAD (<https://osf.io/89kps/>). The complete codebase, pipeline
363 scripts, evaluation artifacts, and security reports are available in the project
364 repository (<https://github.com/DicksonLegend/MedGateRAG>) and will be
365 archived on Zenodo with a persistent DOI upon acceptance.

366 **Declaration of Competing Interests**

367 The authors declare that they have no known competing financial inter-
368 ests or personal relationships that could have appeared to influence the work
369 reported in this paper.

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373 **Generative AI Disclosure**

374 During preparation, the authors used Antigravity (Google’s agentic cod-
375 ing assistant) for code generation and iterative debugging, under direct au-
376 thor supervision. All generated code and outputs were independently verified.
377 The authors authored all manuscript text.

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466 **Appendix A. Hyperparameters, Edge Weights, and System Con-**
467 **figuration**

468 **System Hyperparameters and Execution Environment (Supplemen-**
469 **tary Table S8).** Table A.1 enumerates the complete hyperparameter con-
470 figuration and hardware specifications governing all reported experiments.

471 **Table A.1.** Configuration parameters and computational execution
472 environment.

Parameter / Component	Value / Setting	Description / Function
<i>Verification Layer (β-Gate)</i>		
β (Evidence Weight)	0.70	Balanced hybrid verification (M1: 1.0; M2: 0.0)
τ^* (M1 / M2 / M3 / M4)	0.43 / 0.80 / 0.47 / 0.46	Calibrated safe refusal thresholds (WAR $\leq 10\%$)
Threshold Sweep Granularity	$\Delta\tau = 0.05$	15 discrete points across [0.10, 0.80]
LLM Inference Temperature	$T = 0.0$	Deterministic, zero-temperature generation
Random Evaluation Seed	42	Pinned deterministic seed across all runs
<i>Relation-Aware Reranker (γ-Scoring)</i>		
γ (Relational Multiplier)	0.15	Balanced hybrid reranking (ablation: {0.0, 0.15, 1.0})
DRUG_TREATS / DRUG_CAUSES	1.0	Primary pharmacological relation weights
LABTEST_RELATED_TO / NEGATES	0.8	Diagnostic and negation edge weights
TEMPORAL_BEFORE	0.6	Chronological sequencing edge weight
IS_A / RELATED_TO	0.3	Ontological taxonomic edge weights
<i>Multi-Stream Retrieval Infrastructure</i>		
Dense Retrieval Candidate Pool	$k = 200$	MedCPT + FAISS IVF-PQ candidate pool
Lexical Retrieval Candidate Pool	$k = 200$	BM25 lexical candidate pool
Reciprocal Rank Fusion Constant	$k_{\text{rrf}} = 60$	Standard rank smoothing constant
Top- k Candidates Delivered	$k = 10$	Final passages passed to verification layer
Graph Neighborhood Search Depth	≤ 2 hops	Maximum Kuzu property graph traversal radius
<i>Host Hardware and Memory Footprint</i>		
Central Processing Unit (CPU)	AMD Ryzen 7 7840HS	8 physical cores, 16 threads @ 3.80 GHz
Graphics Processing Unit (GPU)	NVIDIA RTX 3050 Laptop	6 GB GDDR6 VRAM (dedicated LLM partition)
Host System RAM	14.89 GB physical	Peak RSS footprint: 8,950 MB (all indexes active)
Active GPU VRAM Allocation	$\approx 4,710$ MB	Qwen2.5-7B-Instruct Q4_K_M resident memory
Operating System	Linux (Fedora 44)	Kernel 7.1.4-204.fc44.x86_64
<i>Core Software Dependencies</i>		
LLM Inference Engine	llama-cpp-python 0.3.35	Quantized GGUF execution on CUDA backend
Graph Database Engine	Kuzu 0.11.3	In-process property graph database
Vector Search Engine	FAISS 1.15.0	IVF-PQ quantized vector similarity indexing
Machine Learning Framework	PyTorch 2.6.0+cu124	MedCPT encoder inference runtime

Retrieval Drift Invariance Signature. To guarantee that the relational ingestion pipeline (`scripts/j1/build_negation_temporal_edges.py`) did not alter dense vector retrieval behaviour, we computed the SHA-256 hash of the retrieved passage identifiers and scores across golden benchmark queries P01–P05:

- **Pre-Ingestion Signature:**

480 ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac

481 • **Post-Ingestion Signature:**

482 ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e47c9c0d18eac

483 The exact equality of these cryptographic signatures confirms 0.00% retrieval
484 drift (100.0% byte-identical retrieval preservation) following graph enrich-
485 ment.